Department of Information Systems and Technologies

CTIS221 – Object Oriented Programming

FALL 2026 - 2027

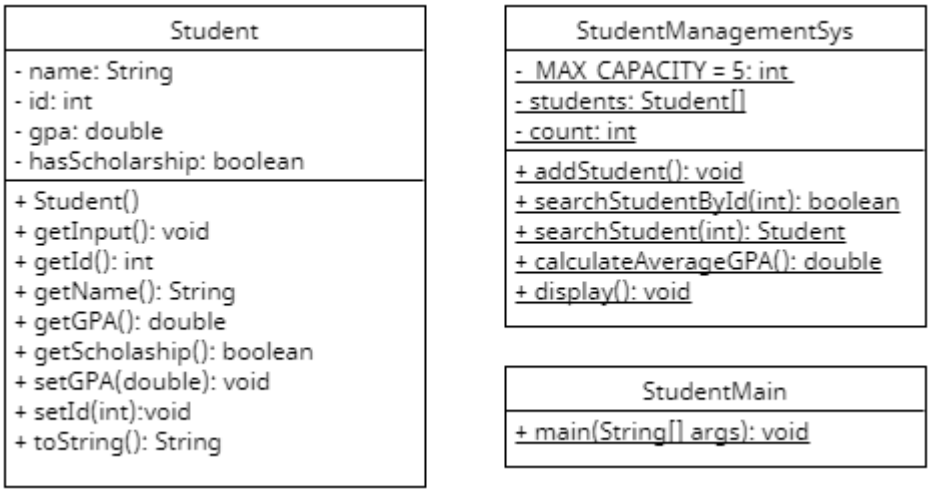
Lab Guide #9

OBJECTIVES: Object Methods, Array of objects, System Classes
Instructors : Neşe Şahin Özçelik **Assistants** : Berk ÖNDER, Efe Mert ŞAHİNKOÇ

Q1. Write a Java program that manages and displays information of **Student** objects. Create a **Student** class structure with the given data members. Also, implement a no-arg constructor, necessary getters & setters, and a **toString**() method.

Then create a **StudentManagementSys** class that has a static data member of type int named **MAX_CAPACITY** (constant), a static data member of type Student[] named **students**, and a static integer **count** data member. Also, implement the static methods **addStudent()**, **searchStudentById()**, **searchStudent(), calculateAverageGPA()**, and **display()**.

Finally, create a **StudentMain** class that has the static **main()** method.



Create a **Student** class with the following instructions:

- Implement a **getInput**() method that gets the data member values from the user. The method should ask the user if the student is a scholarship holder, taking Y/y or N/n as input and setting the boolean value of hasScholarship accordingly.
- Implement necessary getter & setter methods.
- Implement a **toString**() method. The method should check whether student has scholarship or not. Display scholarship status accordingly.

Create a **StudentManagementSys** class with the following instructions:

- Declare a static final member **MAX_CAPACITY** to define the maximum size of the **students** array. The value will be 5.
- Declare and create an object array **students** of type **Student[]** with **MAX_CAPACITY** constant value (5) as its size.
- Declare a static member count to store the number of students.
- Implement an **addStudent()** method that gets student information by invoking the **getInput()** method for each **Student** object. It should check whether a student ID is already assigned by invoking the **searchStudentById()** method and store the object in the **students** array if the ID is available. Also, ask the user if they want to continue adding new students (Y/N: Y for yes, N for no). If the user enters 'N', the loop terminates. The method also checks whether **MAX_CAPACITY** has been reached or not.
- Implement a **searchStudentById()** method that takes a student ID as a parameter, searches the ID in each Student object in the **students** array, and returns true if the ID is available, otherwise false.
- Implement a **searchStudent()** method that takes the **id** of the student as a parameter and searches for the student **id** in each **students** array. If found, return the object; else, return null.
- Implement a **display()** method to display the objects in the **Students** array.
- Implement a **calculateAverageGPA()** method in the **StudentManagementSys** class that loops through the **Students** array and calculates the average GPA of all students in the array.

- It should return the average GPA value as a double.
- If no students are present, it should return 0.

Create a **StudentMain** class with the following instructions:

- Input 5 Student objects by invoking the **addStudent()** method of the **StudentManagementSys** class.
- Try to input a **Student** object that has the same student ID as one of the student objects already in the **Student** array.
- Try to add 2 more **Student** objects to exceed **MAX_CAPACITY**.
- Display all students by invoking the **display()** method of the **StudentManagementSys** class.
- Calculate the average GPA of students in the array by invoking the **calculateAverageGPA()** method of the **StudentManagementSys** class, and display the average GPA.
- Ask the user which student to search by ID, and then display it using the **searchStudent()** method of the **StudentManagementSys** class.

Output:

```
Enter Student Id: 1
Enter Student Name: Efe
Enter Student Gpa: 3,33
Does student has scholarship? (Y/N): Y
Student added successfully!
Do you want to add another student? (Y/N): Y
Enter Student Id: 2
Enter Student Name: Ege
Enter Student Gpa: 2,22
Does student has scholarship? (Y/N): Y
Student added successfully!
Do you want to add another student? (Y/N): Y
Enter Student Id: 3
Enter Student Name: Ugur
Enter Student Gpa: 1,11
Does student has scholarship? (Y/N): N
Student added successfully!
Do you want to add another student? (Y/N): Y
Enter Student Id: 3
Student ID already exists. Please enter a different ID.
Enter Student Id: 4
Enter Student Name: Diana
Enter Student Gpa: 3,45
Does student has scholarship? (Y/N): Y
Student added successfully!
Do you want to add another student? (Y/N): Y
Enter Student Id: 5
Enter Student Name: Ezgi
Enter Student Gpa: 2,32
Does student has scholarship? (Y/N): Y
Student added successfully!
Do you want to add another student? (Y/N): Y
Maximum capacity of students reached.
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Here is a list of the Students:
Student Name=Efe,
 Id=1,
 Gpa=3.33,
 Scholarship Status=Student has scholarship.

Student Name=Ege,
 Id=2,
 Gpa=2.22,
 Scholarship Status=Student has scholarship.

Student Name=Ugur,
 Id=3,
 Gpa=1.11,
 Scholarship Status=No scholarship.

Student Name=Diana,
 Id=4,
 Gpa=3.45,
```

```
 Scholarship Status=Student has scholarship.

Student Name=Ezgi,
 Id=5,
 Gpa=2.32,
 Scholarship Status=Student has scholarship.



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Avg of all Student GPA's: 2,49
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Which student would you like to search? Enter his/her id: 1

Here is the student you were looking for...
Student Name=Efe,
 Id=1,
 Gpa=3.33,
 Scholarship Status=Student has scholarship.
```